Returaj Burnwal

Indian Institute of Technology Madras
returaj.burnwal@gmail.com | https://returaj.github.io | Google Scholar

Education

Indian Institute of Technology Madras *Chennai, India*
PhD Computer Science and Engineering *2022 - present*
- GPA: 9.4/10
- Advisor: Prof. B. Ravindran, Prof. Nirav P. Bhatt

Indian Institute of Technology Hyderabad *Hyderabad, India*
BTech Electrical Engineering *2012 - 2016*
- GPA: 9.16 / 10
- Undergrad research advisor: Prof. Soumya Jana

Professional Experience

2023 **Sakura Science Program**, University of Tokyo, Japan
2022-2023 **Student Researcher**, Google Research, India
2021-2022 **Post Baccalaureate**, Indian Institute of Technology Madras, India
2016-2020 **Software Engineer**, Yahoo!Japan, Japan
2015 **Research Intern**, Uurmi Systems, India

Publications

Peer-Reviewed Articles

OSIL: Learning Offline Safe Imitation Policies with Safety Inferred from Non-preferred Trajectories.
R Burnwal, NP Bhatt, B Ravindran
Autonomous Agents and Multi-Agent Systems (AAMAS), 2026

SafeMIL: Learning Offline Safe Imitation Policy from Non-Preferred Trajectories.
R Burnwal, NP Bhatt, B Ravindran
Association for the Advancement of Artificial Intelligence (AAAI), 2026

GAN-MPC: Training Model Predictive Controllers with Parameterized Cost Functions using Demonstrations from Non-Identical Experts.
R Burnwal, A Santara, NP Bhatt, B Ravindran, G Aggarwal
Robotics: Science and Systems (RSS) Workshop, 2023 (Spotlight)

In Review

NOPE: Can Agents Learn Safe Behavior From Non-Preferred Demonstrations?
OV. Mahajan†, D. Venula†, S. Kuppurajan†, **R Burnwal**, NP Bhatt, B Ravindran, 2026

Learning from Observation: A Survey of Recent Advances
R Burnwal, H Mehta, NP Bhatt, B Ravindran, 2025

Presentations

Invited Talks & Posters

2024. Wadhwani School of Data Science and AI, IIT Madras, India. SafeMIL: Learning Offline Safe Imitation Policy.

2023. Amazon Research Days, Bangalore, India. GAN-MPC: Learning from Non-Identical Experts.

2023. 3rd RBCDSAI Conference of Deployable AI, India. GAN-MPC: Learning from Non-Identical Experts.

Awards & Fellowships

2026 **Student Travel Grant**, AAAI
2023 **Teaching Assistance Award**, Department of Computer Science, IIT Madras
2023 **Prime Minister Research Fellowship**, Ministry of Education, India
2016 **3rd in Hackday**, Yahoo! Japan
2014 **Academic Excellence Award**, IIT Hyderabad
2012-2016 **Merit-Cum Mean Scholarship**, (Fully covers academic fees), IIT Hyderabad
2011 **2nd in Regional Mathematical Olympiad**, Jharkhand, India

Mentoring

2025- **Adithyaa RG**, BTech IIT Madras, Heterogeneous Imitation Learning
2024-2025 **Hriday Mehta**, BTech NIT Surathkal, LfO Survey Paper

Teaching Experience

2025-2023 **Reinforcement Learning**, Graduate Teaching Assistant
2022 **Big Data Laboratory**, Graduate Teaching Assistant
2022 **Problem Solving using Computers**, Undergrad Teaching Assistant

Outreach & Other Responsibilities

2014-2015 **Head Coordinator Robotics Club**, IIT Hyderabad